ConsentChain: Contract-Enforced Dynamic Consent and Verifiable Federated Unlearning for IoT on PureChain

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***Abstract*—Federated learning over consumer IoT must now honour data-subject withdrawal under GDPR, the EU AI Act, and CCPA: consent can be revoked and the trained model must demonstrably forget. Prior work treats consent management, machine unlearning, and on-chain auditability as disjoint threads. ConsentChain closes this *consent–unlearning–verification triangle* in one chain-enforced pipeline: a Consent Lifecycle State Machine as soulbound ERC-721 tokens, a Verifiable Unlearning Protocol gating on-chain acceptance on three independent membership-inference attacks, and a three-contract Solidity architecture on gas-free PureChain (PoA^2) and gas-based Amoy (PoS). Across N-BaIoT and CASAS, per-subject lifecycle cost is $0 on PureChain versus $0.00120 on Amoy, and the gate accepts approximate unlearning (MIA < 0.55) yet rejects naive retraining (whose distributional uniqueness inflates MIA) and any attestation lacking a 3-of-5 verifier threshold, staying faithful to MIA evidence rather than the controller's claim.**

***Index Terms*—federated learning, machine unlearning, smart contracts, dynamic consent, edge intelligence, trustworthy AI, IoT, GDPR, EU AI Act, membership inference attack**

I. INTRODUCTION

Federated learning (FL) trains a shared model without centralising client data [1], often the only legally viable option in consumer IoT [2]. But GDPR Articles 7 and 17 demand purpose-granular, revocable consent whose withdrawal propagates into the model. Federated unlearning (FUL) erases a client's data and influence, yet a bare withdrawal is a one-shot trigger, not a scoped state binding round selection, so training on revoked data may continue unrecorded; and approximate unlearning, the only kind tractable at FL scale, offers no guarantee, fails *silently*, and is self-reported by the controller that owes it. ConsentChain couples *dynamic consent* with *verifiable machine unlearning* in one chain-enforced protocol, closing both gaps.

ConsentChain makes four contributions: (1) a Consent Lifecycle State Machine (CLSM), a soulbound-ERC-721 automaton of six states and nine transitions whose safety and liveness hold by construction; (2) a Verifiable Unlearning Protocol (VUP), method-agnostic, accepting a UA iff three *independent* MIAs (shadow [3], LiRA [4], label-only [5]) fall below τ under t-of-n ECDSA signatures; (3) three Solidity contracts (ConsentRegistry, UnlearningOracle, FLGatekeeper) on both a gas-free (PureChain, PoA^2) and gas-based (Amoy, PoS) chain, the first side-by-side cost comparison; and (4) a dual-domain evaluation (9-client N-BaIoT [6] autoencoder, 5-client CASAS BiLSTM) over three approximate-unlearning methods plus exact retraining, measuring forgetting, cost, and utility recovery.

II. RELATED WORK

Three threads each solve part of the problem but none their conjunction. MedRec [7] records consent immutably yet treats the record as an endpoint: no training is gated by it. Federated-unlearning methods, SISA [8] (shard retraining), FedEraser [9] (history replay), and gradient ascent, remove a contribution but, like the broader survey [10], never specify how unlearning is triggered or bound back to consent. Blockchain-FL brings computation on-chain without consent: BlockFL [11] and BFLC [12] log updates yet let no participant withdraw, and the closest work, Zuo et al. [13], certifies that an unlearning routine *ran*, not that the model *forgot*, the very gap membership inference [3]–[5] measures (a single attack can overstate it [14], motivating our three-attack conjunction).

TABLE I
PRIOR WORK ACROSS THE FOUR CONTRIBUTION DIMENSIONS PLUS COST. *Consent* = GATES TRAINING (NOT MERELY RECORDED); *Verify* = MEASURES FORGETTING (NOT MERELY RECORDS THE ACTION); *Partial* = RECORDED BUT NOT ENFORCED OR MEASURED.

System	Consent	Unlearn	Verify	On-chain	Gas-free
MedRec [7]	Partial	No	No	Partial	No
SISA/FedEraser [8], [9]	No	Yes	No	No	n/a
BlockFL/BFLC [11], [12]	No	No	No	Yes	No
Zuo et al. [13]	No	Yes	Partial	Yes	No
ConsentChain (ours)	Yes	Yes	Yes	Yes	Yes

Each prior system covers at most one dimension; ConsentChain supplies the missing edges (Table I): consent state gating rounds under CLSM properties, any unlearning paradigm as a black box, and three-MIA verification driving the consent transition. Chain choice is economic, not protocol: identical contracts run on PureChain [15], a public gas-free PoA^2 chain, and on gas-based Amoy, a difference quantified in Section IV.

III. SYSTEM MODEL AND FRAMEWORK

A. System model

Six entities $\langle DS_i, DC, \{FL_k\}, VC, CR, UO, GK\rangle$ compose ConsentChain: data subject DS_i (key-controlled owner of one soulbound Consent NFT), controller DC (drives

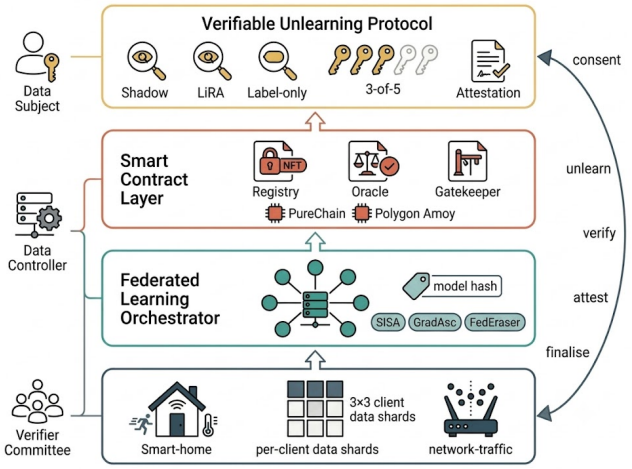


Fig. 1. ConsentChain architecture: off-chain IoT data and Flower FedAvg layers, the three on-chain Solidity contracts on PureChain / Polygon Amoy, and the off-chain Verifiable Unlearning Protocol. The clockwise loop consent→unlearn→verify →attest→finalise is the edge ConsentChain closes.

the round loop, triggers unlearning on revocation), devices FL_k (local SGD), Verifier Committee $VC = \{v_1, \ldots, v_n\}$ ($n = 5$ ECDSA keys, threshold $t = 3$), and the three contracts `ConsentRegistry` (CR), `UnlearningOracle` (UO), `FLGatekeeper` (GK); any chain reader is an *auditor*. ConsentChain runs FedAvg [1],

$$\theta^{(r+1)} = \sum_{k \in K_r} \frac{n_k}{\sum_{j \in K_r} n_j} \theta_k^{(r+1)}, \tag{1}$$

with the participant set K_r in (1) fixed to GK.`validateRound`'s return value, not an orchestrator-chosen subset: consent is enforced at the membership, not the gradient, level.

B. Consent Lifecycle State Machine

The CLSM is a deterministic finite automaton CLSM $= (S, \Sigma, \delta, s_0, F)$ with start $s_0 = \textsc{Ina}$, absorbing terminal $F = \{\textsc{Unl}\}$, and states $S = \{\textsc{Ina}, \textsc{Act}, \textsc{Sus}, \textsc{Rev}, \textsc{Exp}, \textsc{Unl}\}$ (inactive, active, suspended, revoked, expired, unlearned). Its nine δ-edges over the eight events Σ are

$$\begin{aligned} &\texttt{grant}: \textsc{Ina} \to \textsc{Act}, \quad \{\texttt{modify}, \texttt{tick}\}: \textsc{Act} \circlearrowleft, \\ &\texttt{revoke}: \textsc{Act} \to \textsc{Rev}, \quad \texttt{expire}: \textsc{Act} \to \textsc{Exp}, \\ &\texttt{suspend/resume}: \textsc{Act} \leftrightarrow \textsc{Sus}, \\ &\texttt{unlearn_verified}: \{\textsc{Rev}, \textsc{Exp}\} \to \textsc{Unl}. \end{aligned} \tag{2}$$

Safety is immediate from (2): no edge targets ACT from $\{\textsc{Rev}, \textsc{Exp}\}$, `unlearn_verified` is oracle-only, and the lone `validateRound` edge `tick` guards on $\rho_r \in \mathcal{P}$. `ConsentRegistry` realises δ one-edge-per-function; the soulbound constraint (no `transferFrom`) confines token-i transitions to DS_i.

C. Verifiable Unlearning Protocol

On $\text{state}(i) \to \{\textsc{Rev}, \textsc{Exp}\}$ the gatekeeper sets `pendingUnlearningSince`$[i] = b$ and `validateRound` blocks all rounds (liveness). Within $\Delta_{\max}$ blocks the controller must unlearn, score f_{post} with the three MIAs, assemble the Unlearning Attestation, gather t-of-n signatures, and submit.

Unlearning methods. ConsentChain is method-agnostic over the three families of Section II: SISA-Fed [8] (per-shard retrain), gradient ascent ($\theta^{(j+1)} = \theta^{(j)} + \eta \nabla \mathcal{L}(\theta^{(j)}; \mathcal{D}_i)$, $k = 10$ steps, $\eta = 10^{-4}$), and FedEraser [9] (history replay with 3-round calibration).

Membership-inference attacks. Against DS_i's set $\mathcal{D}_i$ (members M), three independent attacks each score f_{post} in $[0, 1]$ (higher = more residual membership). The *shadow-model* attack [3], adapted to reconstruction models [16], trains an MLP classifier on the "in"/"out" reconstruction errors of $S = 5$ shadow autoencoders; s_{sh} = the fraction of M it flags "in." *LiRA* [4], the strongest, fits in-/out-pool Gaussians to the log-reconstruction-errors of $n_{\text{in}} = n_{\text{out}} = 4$ reference models per $x \in M$ and takes s_{lira} = TPR at an α-controlled FPR ($\alpha = 0.05$; $\alpha = 0.5$ ablation in Section V). The *label-only* attack [5], seeing only the reconstruction decision, thresholds at FPR 0.05 and takes s_{lo} = the fraction of M below it.

Attestation and hash binding. The Unlearning Attestation is the tuple $u = \langle i, h_{\text{pre}}, h_{\text{post}}, m, s_{\text{sh}}^{\text{fp}}, s_{\text{lira}}^{\text{fp}}, s_{\text{lo}}^{\text{fp}}, t_u \rangle$: token ID, pre-/post-unlearning model hashes, method $m \in \{0, 1, 2\}$, fixed-point scores $s_j^{\text{fp}} = \lfloor 10^4 s_j \rfloor$, timestamp. Each verifier signs $h_u = \text{keccak256}(c \,\|\, u)$ over $c =$ `block.chainid`, binding the UA to one chain against cross-chain replay. Writing $R(\Sigma)$ for the distinct registered signers of h_u and $h_{\text{rev}}(i) =$ CR.`modelHashAtRevocation`(i), the oracle finalises iff

$$\begin{aligned} \text{accept}(u, \Sigma): \Big(\textstyle\bigwedge_j s_j^{\text{fp}} < 5500\Big) \wedge |R(\Sigma)| \geq t \\ \wedge\ h_{\text{pre}} = h_{\text{rev}}(i) \wedge \text{state}(i) \in \{\textsc{Rev}, \textsc{Exp}\}. \end{aligned} \tag{3}$$

The last two conjuncts pin the UA to the exact model version and block re-attestation of an already-UNL token, closing the replay loop. With $\tau^{\text{fp}} = 5500$ just above the 0.5 random-guess baseline (margin for finite-$|M|$ noise), the three-attack conjunction is what makes so tight a threshold tolerable.

D. Three-contract Solidity architecture

ConsentChain is three Solidity 0.8.24 contracts (NatSpec in supplementary `docs/contracts-spec.md`). `ConsentRegistry`, the soulbound ERC-721 realising δ above, adds a write-once h_{rev} and a `transitionToUnlearned` callable only by the oracle. `UnlearningOracle` holds VC, t, and τ^{fp}; `submitAttestation` evaluates $\text{accept}(u, \Sigma)$ and, on success, atomically finalises the token and unblocks the gatekeeper. `FLGatekeeper` gates rounds: `validateRound` reverts on any ineligible participant or pending unlearning (encoding liveness), and permissionless `reportTimeout` emits `ComplianceViolation` past the $\Delta_{\max}$ deadline. The loop closes when a revocation fires `recordPendingUnlearning` and only a valid UA, via the oracle's `unblock`, releases the gatekeeper (Figure 1).

E. Threat model

We defend four in-scope adversaries: a *lazy orchestrator* (GK reverts on pending unlearning, emits `ComplianceViolation` after $\Delta_{\max}$), a *deceptive controller* (three MIAs plus VC t-of-n signatures), a *consent forger* (soulbound NFT, `dsOwner` check), and a *replay attacker* (UA binding (c, i, h_{pre})). The committee is the trust floor: $\geq t$ colluding verifiers can certify an invalid attestation. Our journal remedy stakes verifiers, slashable on a successful challenge: DS_i, the one party holding $\mathcal{D}_i$ and motivated to dispute, contests an accepted UA by exhibiting a contradicting MIA evaluation against the attested $(h_{\text{pre}}, h_{\text{post}})$, with per-attestation committees VRF-drawn from a larger pool against pre-collusion. Colluding participants (an orthogonal DP layer) and six further classes are out of scope (Section V); the full per-adversary model is in supplementary `docs/threat-model.md`.

F. Formal properties

ConsentChain satisfies four properties: the first pair by construction, the second empirically (Section IV). *Safety (consent integrity):* `validateRound` admits only ACT tokens with $\rho_r \in \mathcal{P}_k$, and no δ-path re-enters ACT (above). *Audit completeness:* every transition, attestation, violation, and model-hash update emits an indexed event. *Enforceability (liveness):* within $\Delta_{\max}$ blocks of a revocation, either a valid UA drives the token to UNL or the gatekeeper logs a public `ComplianceViolation`, degrading to "no training," not "unconsented training." *Verifiability:* the oracle finalises iff $\text{accept}(u, \Sigma)$ (3) holds. Mechanised proofs are left to future work.

IV. EVALUATION

We evaluate unlearning effectiveness, on-chain cost (PureChain versus Polygon Amoy), round-validation scalability to $N = 500$, and utility recovery, plus two on-chain demos of the liveness and verifiability properties; the reproducibility recipe is in supplementary `docs/eval-protocol.md`.

A. Experimental setup

The primary federation is a 9-client N-BaIoT anomaly detector [6], one autoencoder per device, a symmetric MLP $115 \to 75 \to 50 \to 25 \to 10$ trained with FedAvg for 30 rounds (Flower, seed 42). A GDPR Art. 17 erasure of client 5 at round 30 triggers one of the three approximate methods or exact 8-client retraining. Contracts run live on gas-free PureChain (PoA^2) and gas-based Amoy (PoS), with VC of five ECDSA keys, $t = 3$, $\Delta_{\max} = 1000$ blocks.

B. Unlearning effectiveness

Table II reports each method's three MIA scores on the revoked client; the `submitAttestation` predicate accepts iff all three sit below $\tau = 0.55$.

The gate accepts all three approximate methods — shadow MIA $0.438/0.254/0.523$ (gradient ascent/SISA-Fed/FedEraser), LiRA and label-only far below τ. Strikingly, it *rejects* exact naive retraining, which never trained on client 5: shadow scores $0.584 > \tau$ because it reads client 5's distinctive traffic distribution, not membership [3], a confound LiRA (0.045) and label-only (0.014) sidestep. The no-unlearning baseline already sits near random $(0.55/0.05/0.01)$, so this federation memorises client 5 only weakly: here the gate's role is rejecting that distributional confound, not detecting strong leakage. Section IV-E supplies the high-memorisation regime where it must, and does, reject real residual signal. All three accepted UAs finalise end-to-end on PureChain to UNLEARNED.

TABLE II
UNLEARNING EFFECTIVENESS: MIA SCORES ON THE REVOKED CLIENT'S DATA, POST-UNLEARNING. THE GATE ACCEPTS IFF ALL THREE ARE BELOW $\tau = 0.55$.

Method	Shadow MIA	LiRA	Label-only	On-chain accept
No unlearning (baseline)	0.5455	0.0475	0.0060	Yes
Gradient ascent	0.4380	0.0435	0.0030	Yes
SISA-Fed	0.2542	0.0795	0.0862	Yes
FedEraser	0.5232	0.0325	0.0050	Yes
Naive retrain (exact ref.)	0.5843	0.0445	0.0138	No

C. On-chain cost and scalability

Table III reports per-operation gas and cost from a lifecycle $\times 10$ run (USD at POL/USD $= 0.0953$).

TABLE III
PER-OPERATION GAS AND COST, LIFECYCLE $\times 10$ MEAN. PER-DS LIFECYCLE = SUM OF THE FOUR OPS.

Operation	PureChain gas	Amoy gas	Amoy MATIC	Amoy USD
`updateModelHash`	30,074	32,673	0.0010	\$0.00009
`grantConsent`	144,868	153,568	0.0046	\$0.00044
`revokeConsent`	107,904	96,404	0.0029	\$0.00028
`submitAttestation` (3-of-5)	102,903	137,600	0.0041	\$0.00039
Per-DS lifecycle	**385,750**	**420,245**	**0.0126**	**\$0.00120**

Per-subject lifecycle cost is \$0 on PureChain and \$0.00120 on Amoy; the gap is gas *price* (zero versus $\sim$ 30 gwei), not count. `validateRound`(N) scales exactly linearly: PureChain runs the full $N \in [5, 500]$ sweep live ($R^2 \approx 1$), while Amoy, measured at $N \in \{5, 10\}$ and extrapolated, projects to \$0.0176 per $N = 500$ round, i.e. \$1.76/day at 100 rounds/day versus \$0; off-chain, `validateRound` adds only $\approx$3.2 ms per round at $N = 500$. Gas-free infrastructure is thus a different cost class, not an optimisation. Two short PureChain runs confirm the remaining properties: a $\Delta_{\max} = 2$ timeout fires `ComplianceViolation` (liveness), and a t-of-5 sweep exercises both branches of the acceptance predicate (verifiability).

D. Utility recovery after revocation

Resuming FL for ten rounds after dropping client 5, the post-revoke model reaches reconstruction error 0.4720, 6.1% *better* than the 9-client baseline (0.5025), since client 5 was an outlier. Gradient ascent (0.4726) and FedEraser (0.4698) match this at no utility cost, while SISA-Fed pays $+29\%$ $(0.6081$ by round 40), consistent with its stronger forgetting (Figure 2).

E. Cross-domain validation: the high-memorisation regime

CASAS provides the complementary high-memorisation regime: where N-BaIoT's baselines sit near random, the 5-client CASAS BiLSTM reaches pre-unlearning shadow MIA

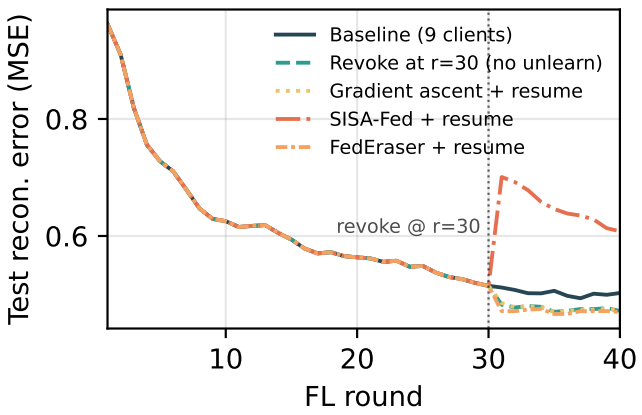


Fig. 2. Utility recovery. Centralized test reconstruction error per FL round. Trajectories diverge at round 30: SISA-Fed pays a real utility cost, gradient ascent and FedEraser converge to within 0.5% of plain post-revoke by round 40.

0.979, so the gate must *reject*, and does: gradient ascent (0.980), FedEraser (0.840), and even exact naive retrain (0.846) stay above τ and are refused, while only SISA-Fed (0.499/0.035/0.000) clears all three attacks. The two domains thus exercise both branches of the acceptance predicate on identical contracts; full results appear in the journal version.

V. Discussion and Conclusion

Our gate is deliberately conservative: shadow MIA conflates distributional uniqueness with membership (naive retrain, never trained on the data, scored 0.584/0.846), so requiring *all three* MIAs below τ errs toward false rejection, and LiRA, which resists the confound but is α-sensitive, is co-calibrated with τ per model class. ConsentChain targets *approximate* unlearning, meeting GDPR Art. 17's "reasonable measures" by predicate plus on-chain audit rather than formal erasure: certified erasure stays intractable for approximate methods at FL scale, so the gate is evidence-based by design. Formal-erasure and ECDSA-leakage guarantees are out of scope (both BLS-collapsed in the journal), as is preventing off-chain training on revoked data *before* a round runs — a controller that does so still cannot produce a passing attestation, so the omission is publicly detectable. Our federations are dataset-bound — nine N-BaIoT device traces, five CASAS apartments — while the on-chain path scales independently to $N = 500$; carrying the learning and MIA side to hundreds of clients is journal work. Its consent, liveness/verifiability, and audit mechanisms discharge GDPR Arts. 7/17/21, EU AI Act Arts. 15/26, and CCPA § 1798.105, and run unchanged on PoA^2 or permissionless Amoy: the economic model, not the protocol, is the deployment variable.

ConsentChain is, to our knowledge, the first protocol binding dynamic consent, machine unlearning, and on-chain verification in one chain-enforced pipeline, closing a triangle prior work addressed only at the vertices. Across 9-client N-BaIoT [6] and 5-client CASAS, approximate unlearning drives all three MIAs below $\tau = 0.55$ at $\leq 3\%$ utility loss, with per-subject lifecycle cost \$0 on PureChain versus \$0.00120 on Amoy. Beyond the BLS aggregation and formal-erasure guarantees noted above, future work targets zero-knowledge eligibility proofs, PoA^2-native verifiers, and cross-chain consent portability.

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